

■ Authentication Security Incident Report

Investigation Type: Synthetic / Educational

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1. Executive Summary

A review of a synthetic authentication dataset identified several suspicious authentication patterns. The dataset contains evidence consistent with password spraying against multiple accounts, repeated authentication failures against a single account, and successful authentication followed by privileged access. The investigation separates directly observed events from hypotheses requiring additional evidence.

2. Scope

The investigation covered authentication and access events in the synthetic log. Fields analyzed included timestamp, source IP, username, action, result, and device.

3. Detection Findings

Finding 1 — Password Spraying Pattern

Source IP 185.220.101.10 attempted authentication against five different accounts within a short period. Targeted accounts were alice, bob, charlie, david, and emma. This behavior is consistent with a password-spraying pattern.

Finding 2 — Repeated Authentication Failures

Source IP 10.10.10.50 generated five failed authentication attempts against the alice account. A successful authentication occurred shortly afterward.

Finding 3 — Unusual Privileged Access

After the successful authentication, alice accessed VPN resources and subsequently accessed the admin panel. The log establishes the sequence but does not independently prove account compromise or unauthorized activity.

Finding 4 — Service Account Activity

The service_account successfully authenticated from 203.0.113.45 and subsequently accessed database administration and configuration resources. Further investigation is required to determine whether this activity was authorized.

4. Incident Timeline

Time (UTC)	Event
08:15:01	Authentication failures begin against multiple accounts
08:15–08:16	Multiple accounts targeted from 185.220.101.10
08:20:02	Repeated failures begin against alice
08:20:10	Fifth failed authentication attempt

Time (UTC)	Event
08:21:02	Successful authentication as alice
08:22:15	VPN access
08:24:11	Admin-panel access
08:30:00	Service account successfully authenticates
08:30:20	Database administration access
08:31:02	Configuration access
08:32:10	Service account logs out

5. Evidence vs. Assumptions

Direct Evidence

- Multiple accounts were targeted from 185.220.101.10.
- Five failed login attempts were recorded against alice.
- A successful login occurred after those failures.
- alice accessed VPN resources and the admin panel.
- service_account accessed database administration and configuration resources.

Assumptions / Hypotheses

- The password-spraying activity was malicious.
- The alice account was compromised.
- The admin-panel access was unauthorized.
- The service account was compromised or misused.

These hypotheses require additional evidence before they can be confirmed.

6. Risk Assessment

Risk	Assessment
Password spraying	High
Brute-force activity	High
Successful login following repeated failures	High
Privileged access following suspicious authentication	High
Service-account anomaly	High

7. Recommended Response

Immediate Actions

- Review the alice account.
- Reset credentials if compromise is suspected.
- Revoke active sessions.
- Review MFA activity.

- Investigate admin-panel activity.
- Review service-account activity.
- Preserve relevant authentication and application logs.

Detection Improvements

- Alert when one IP targets multiple accounts.
- Alert when repeated failures are followed by success.
- Monitor privileged access after suspicious authentication.
- Monitor service accounts for unusual activity.
- Centralize authentication logs in a SIEM.

Preventive Controls

- Multi-factor authentication
- Authentication rate limiting and throttling
- Strong password controls
- Least-privilege access
- Service-account restrictions
- Continuous security monitoring

8. Conclusion

The synthetic authentication dataset contains multiple patterns that warrant investigation. The strongest observable indicators are the multi-account authentication attempts from 185.220.101.10, repeated failures against alice, the subsequent successful authentication, and later VPN/admin-panel access. Privileged service-account activity is also present. The available data establishes these events but does not, by itself, prove malicious intent or account compromise. Additional endpoint, VPN, MFA, application, firewall, and identity-provider logs should be reviewed before making a final incident determination.

Note: This report uses synthetic data created for cybersecurity training. No real user credentials or confidential organizational logs are included.